

Notes: Costinot & Vogel (JPE, 2010)

Matching and Inequality in the World Economy

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1 Big picture

- **Question.** How can we analyze changes in factor allocation and inequality when there are many worker types and many tasks rather than the textbook two-factor/two-good case?
- **Core model.** A continuum of workers with skill s is assigned to a continuum of tasks with skill intensity σ .
- **Key simplification.** The marginal product of a worker in a task does not depend on the set of other workers in that task, so the competitive equilibrium becomes a monotone matching problem.
- **Main closed-economy lesson.** Skill abundance, skill diversity, skill-biased technology, and extreme-biased technology each generate distinct matching and inequality patterns.
- **Main trade lesson.** North-South trade driven by factor endowments raises inequality in the skill-abundant country and lowers it in the skill-scarce country; North-South trade driven by technology bias reverses the conclusion.
- **Economic takeaway.** The paper gives a unified assignment-based language for globalization, technological change, job polarization, and full-distribution wage inequality.

2 Incremental contribution of the paper

- It provides robust comparative statics in many-good, many-factor environments where standard trade models often yield weak predictions.
- It introduces nonparametric concepts of skill abundance, skill diversity, skill-biased technology, and extreme-biased technology.
- It predicts changes over the full wage distribution rather than only a skill premium.
- It unifies North-South trade, North-North trade, global SBTC, and offshoring in one assignment framework.
- It explains why empirical effects of globalization can be mixed: factor-endowment and technology-bias channels can move inequality in opposite directions.

3 Model objects and measurement

3.1 Workers, tasks, and final output

Workers are indexed by skill s and supplied according to $V(s)$. Tasks are indexed by skill intensity σ and weighted by $B(\sigma)$.

$$Y = \left\{ \int_{\sigma \in \Sigma} B(\sigma) [Y(\sigma)]^{(\varepsilon-1)/\varepsilon} d\sigma \right\}^{\varepsilon/(\varepsilon-1)}.$$

$$Y(\sigma) = \int_{s \in S} A(s, \sigma) L(s, \sigma) ds.$$

Interpretation: The model is high-dimensional, but the assignment of workers to tasks is summarized by a matching function.

3.2 Comparative advantage

$$A(s', \sigma') A(s, \sigma) > A(s, \sigma') A(s', \sigma) \quad \text{for } s' > s, \sigma' > \sigma.$$

Interpretation: This log-supermodularity assumption means high-skill workers have comparative advantage in high-skill-intensity tasks.

3.3 Matching and wages

Let $M(s)$ be the task skill intensity performed by a worker of skill s .

$$w(s) = p(M(s))A(s, M(s)).$$

Interpretation: Inequality changes when shocks shift $M(s)$ and change the slope of the wage schedule.

4 Models and theoretical specifications

4.1 Closed-economy supply shocks

Skill abundance is defined by monotone likelihood ratio dominance:

$$V'(s')V(s) \geq V(s')V'(s) \quad \text{for all } s' \geq s.$$

If V' is skill abundant relative to V , then

$$M'(s) \leq M(s).$$

Interpretation: More high-skill workers lower the scarcity value of high skill and compress the return to skill.

Skill diversity means relatively more mass in the tails. If V' is more diverse than V , then there is a threshold s^* such that

$$M'(s) \geq M(s) \quad \text{for } s \leq s^*, \quad M'(s) \leq M(s) \quad \text{for } s \geq s^*.$$

Interpretation: This is job convergence and wage polarization within regions of the skill distribution.

4.2 Closed-economy demand shocks

Skill-biased technology is defined by MLR dominance in $B(\sigma)$. If B' is skill-biased relative to B , then

$$M(s) \leq M'(s),$$

and inequality rises pervasively. Extreme-biased technology reallocates workers away from middle tasks and toward extreme tasks.

Interpretation: Skill-biased change raises the return to skill everywhere; extreme-biased change generates polarization.

4.3 World economy results

Proposition 1. If Home is skill abundant relative to Foreign and technologies are common, trade integration raises inequality at Home and lowers inequality abroad.

Proposition 2. If Home's technology is skill-biased relative to Foreign's, trade integration has the opposite inequality effect.

Proposition 3. If Home is more diverse than Foreign, North-North trade produces tail-specific inequality effects and U-shaped or inverted-U-shaped task-employment changes.

Proposition 4. Global SBTC raises within-country and between-country inequality.

Proposition 5. Offshoring induces task upgrading and a pervasive rise in inequality in both countries.

5 Theoretical design / identification logic

This is a theoretical paper, so there is no empirical identification strategy. The identification-style logic is based on isolating primitive changes in factor supply, task demand, integration, and offshoring.

- Skill abundance and skill diversity isolate factor-supply channels.
- Skill-biased and extreme-biased technology isolate factor-demand channels.
- Factor price equalization maps closed-economy comparative statics into world-economy trade results.
- Each proposition is a counterfactual comparison between two equilibria.
- The predictions rely on monotone comparative advantage and stochastic-order restrictions, not on parametric distributions.

Interpretation: The paper identifies mechanisms by construction: if one primitive changes while the others remain fixed, the direction of matching and inequality follows.

6 Figures and result boxes

6.1 Figure 1: Changes in skill abundance and matching

Read: Panel (a) shows a more skill-abundant distribution. Panel (b) shows $M'(s) \leq M(s)$: workers of a given skill perform less skill-intensive tasks.

Economic message: Skill abundance compresses the return to skill because high skill becomes less scarce.

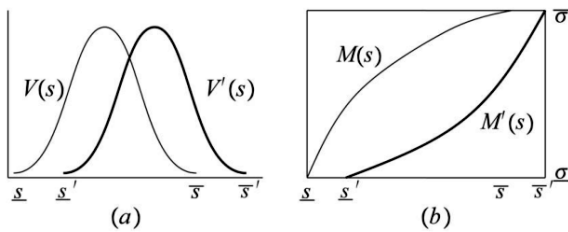


FIG. 1.—Changes in skill abundance and matching

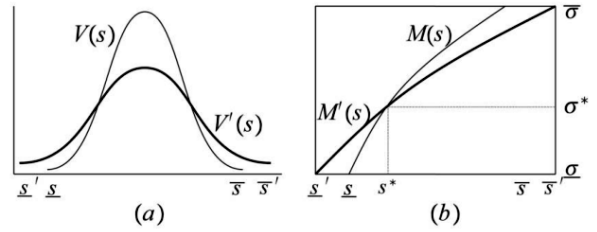


FIG. 2.—Changes in skill diversity and matching

6.2 Figure 2: Changes in skill diversity and matching

Read: Panel (a) shows more mass in the tails. Panel (b) shows a single crossing in matching functions.

Economic message: Skill diversity produces job convergence and tail-specific wage effects.

6.3 Figure 3: Skill- and extreme-biased technological change and matching

Read: Panel (a) shows skill-biased technological change; panel (b) shows extreme-biased technological change.

Economic message: Skill-biased change increases inequality pervasively; extreme-biased change generates job and wage polarization.

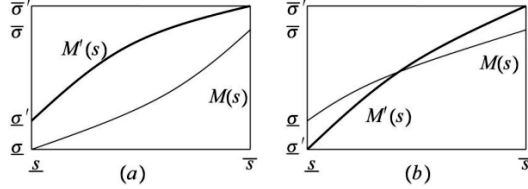


FIG. 3.—Skill- and extreme-biased technological change and matching

Figure 3. Skill- and extreme-biased technological change and matching.

7 Short summary

- The paper builds a many-factor, many-task assignment model.
- Log-supermodularity generates positive assortative matching.
- Supply and demand shocks are described using stochastic-order concepts.
- The model delivers full-distribution inequality predictions.
- North-South trade, North-North trade, SBTC, and offshoring are analyzed in one framework.

8 Conclusion

8.1 Core conclusion

A high-dimensional assignment model can deliver strong and intuitive comparative statics for factor allocation and wage inequality.

8.2 Economic intuition

Inequality changes when shocks move workers along the task ladder. If higher-skill workers move toward tasks where comparative advantage is stronger, the return to skill rises.

8.3 Incremental contribution in one paragraph

The paper extends classic trade-and-inequality reasoning from two factors and two goods to an economy with arbitrarily many workers and tasks. It replaces dimensionality-dependent comparative statics with monotone matching results and predicts changes across the full wage distribution.

8.4 Limitations

The model abstracts from firm heterogeneity, unemployment, bargaining, endogenous education, and team production.

8.5 Takeaway

The paper provides a mathematical language for thinking about globalization and inequality in a world with many worker types and many kinds of work.